# Pöschl–Teller well in a box: closed-form $C_v^{\mathrm{q}}$ and $C_v^{\mathrm{cl}}$

Nadav Jean · meeting of 28.10.2026 · D§n = section $n$ of `PT_Box_Derivation.pdf` (full derivations)

**Assumptions.**

**A1** One particle of mass $m$ in one dimension, in the canonical ensemble at temperature $T$ ($\beta = 1/k_BT$).

**A2** Pöschl–Teller well with $\lambda = 2$. (That it is reflectionless with exactly one bound state follows from $\lambda = 2$; see Sec. 2.)

**A3** Hard walls (infinite square well) at $x = \pm L/2$.

**A4** $\alpha L \gg 1$: terms of order $e^{-\alpha L}$ (the well felt at the walls) are dropped; $< 10^{-40}$ for $\alpha L \geq 100$.

**A5** $L \gtrsim \lambda_{\mathrm{th}}$: the Poisson-summation remainder, of order $e^{-4\pi(L/\lambda_{\mathrm{th}})^2}$, is dropped.

**A6** (Sec. 6 only) Slow strokes (level populations kept) that change $\alpha$ and $L$ with the same $\gamma$.

**Notation.** $a \equiv \beta\hbar^2\alpha^2/m$ (well depth over $k_BT$) and $\lambda_{\mathrm{th}} \equiv \sqrt{2\pi\hbar^2\beta/m}$ (thermal wavelength). **Why $\alpha L$ and not $L$:** the well's only length is its width $1/\alpha$, so $L$ can enter the physics only as $\alpha L$, the box length in well widths. For example $L/\lambda_{\mathrm{th}} = \alpha L/\sqrt{2\pi a}$. Every $C_v$ below depends on $\alpha L$ and $a$ only (D§1).

## 1 Potential

$$V(x) = -\frac{\hbar^2}{2m}\alpha^2\frac{\lambda(\lambda-1)}{\cosh^2(\alpha x)} \xrightarrow{\lambda=2} -\frac{\hbar^2\alpha^2}{m}\frac{1}{\cosh^2(\alpha x)} \quad (|x| < L/2), \qquad V = +\infty \quad (|x| \geq L/2). \tag{1}$$

The well is the "modified Pöschl–Teller well" row of Table I of Ref. [1, p. 021013-9], copied as written there.

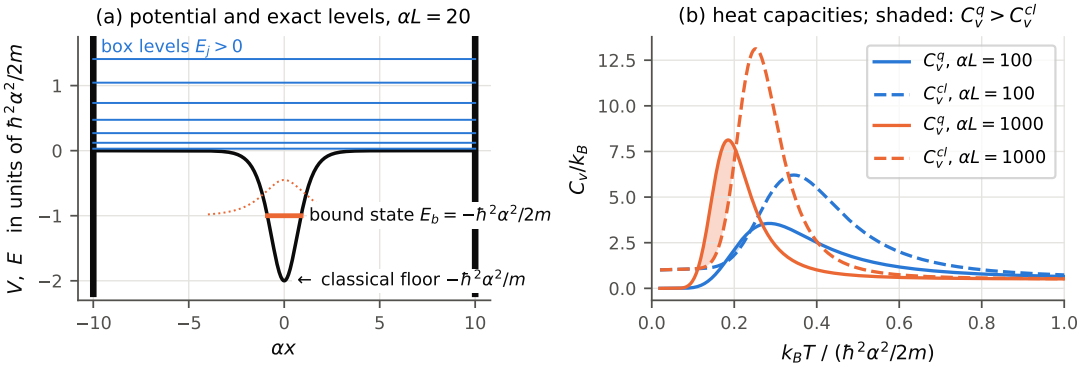



Figure 1: (a) Eq. (1) with its exact levels, $\alpha L = 20$. (b) Eqs. (4) and (7); shaded where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

## 2 Energy levels

$$E_b = -\frac{\hbar^2\alpha^2}{2m}, \qquad E_j = \frac{\hbar^2 k_j^2}{2m}, \quad k_jL = j\pi + 2\arctan\left(\frac{k_j}{\alpha}\right), \quad j = 1, 2, 3, \ldots \tag{2}$$

*Derivation.* With $W = \frac{\hbar\alpha}{\sqrt{2m}}\tanh(\alpha x)$ and $A = \frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W$, one has $A^\dagger A = H_{\mathrm{PT}} + \frac{\hbar^2\alpha^2}{2m}$ and $AA^\dagger = H_{\mathrm{free}} + \frac{\hbar^2\alpha^2}{2m}$ [2, p. 18, Eqs. (32)–(33)]. So $A^\dagger$ maps every free solution to a solution of the well [2, p. 14, Eqs. (8)–(11)]: $\psi_k = (\alpha\tanh\alpha x - ik)\,e^{ikx}$ (D§2.1). Setting $\psi(\pm L/2) = 0$ on the even and odd parts of $\psi_k$ gives the condition for $k_j$ (D§2.3).

*Exactly one bound state.* Partner Hamiltonians have the same spectrum, except that $A^\dagger A$ has one extra level, the zero mode of $A$ [2, p. 15, Eq. (12); p. 17]. The free particle has no bound state, so the well has exactly one: $\psi_b \propto 1/\cosh(\alpha x)$ (which satisfies $A\psi_b = 0$), at $E_b$. Equivalently, $V = -n(n+1)\,\mathrm{sech}^2 x$ "holds $n$ bound states" [2, p. 77, Eq. (312)], with $n(n+1) = \lambda(\lambda-1) = 2$, so $n = 1$; the same page gives $E_1 = -1$ in units $\hbar = 2m = \alpha = 1$. In the box: one even root, no odd root (D§2.4).

## 3 Quantum heat capacity

$$Z_q = e^{-\beta E_b} + \sum_{j\geq 1} e^{-\beta E_j}, \qquad C_v^{\mathrm{q}} = k_B\beta^2\frac{\partial^2 \ln Z_q}{\partial\beta^2} \quad [3, \text{p. 20, Eq. (1.27)}]. \tag{3}$$

*Step 1 (sum → integral).* $\sum_{j\ge1} F(j) = \int_0^\infty F\,\mathrm{d}j - \frac{1}{2}F(0)$, using Poisson summation [4, Eq. (1.8.15)] and A5. Here $F(j) = e^{-\beta E_j}$ is even in $j$, with $F(0) = 1$.

*Step 2 (level density).* From Eq. (2), $\mathrm{d}j/\mathrm{d}k = \frac{1}{\pi}\left[L - \frac{2\alpha}{k^2+\alpha^2}\right]$.

*Step 3 (two integrals).* With $b = \beta\hbar^2/2m$: $\int_0^\infty e^{-bk^2}\mathrm{d}k = \frac{1}{2}\sqrt{\pi/b}$, and $\int_0^\infty \frac{e^{-bk^2}}{k^2+\alpha^2}\mathrm{d}k = \frac{\pi}{2\alpha}e^{b\alpha^2}\operatorname{erfc}(\alpha\sqrt{b})$ [4, Eq. (7.7.1)] (D§3.2–D§3.4). Adding $e^{-\beta E_b} = e^{a/2}$:

$$Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{a/2}\operatorname{erf}\sqrt{a/2}, \qquad \frac{C_v^{\mathrm{q}}}{k_B} = a^2\frac{\partial^2 \ln Z_q}{\partial a^2} \quad \text{(explicit form: D§3.5).} \tag{4}$$

Check: Eq. (4) equals $k_B\beta^2\mathrm{Var}(E)$ summed over the exact levels to $2\times10^{-13}k_B$ (`check_formulas.py`).

## 4 Classical heat capacity

$$Z_{\mathrm{cl}} = \frac{1}{2\pi\hbar}\int\mathrm{d}x\int\mathrm{d}p\, e^{-\beta H} = \frac{1}{\lambda_{\mathrm{th}}}\int_{-L/2}^{L/2} e^{-\beta V}\mathrm{d}x \quad \text{[3, p. 32, Eq. (2.1); p. 35, Eq. (2.6)].} \tag{5}$$

*Step.* Split $\int e^{-\beta V}\mathrm{d}x = L + I$, extend $I$ to all $x$ (A4), substitute $t = \tanh\alpha x$, expand $e^{a(1-t^2)}$ and integrate term by term with the Beta integral [4, Eq. (5.12.1)] (D§4.2):

$$Z_{\mathrm{cl}} = \frac{L+I}{\lambda_{\mathrm{th}}}, \qquad \alpha I = \int_{-1}^{1}\frac{e^{a(1-t^2)}-1}{1-t^2}\,\mathrm{d}t = \sum_{\ell\ge1}\frac{(2a)^\ell}{\ell\,(2\ell-1)!!}, \tag{6}$$

$$\frac{C_v^{\mathrm{cl}}}{k_B} = \frac{1}{2} + a^2\left[\frac{\alpha J_2}{\alpha L+\alpha I} - \left(\frac{\alpha J_1}{\alpha L+\alpha I}\right)^2\right], \quad \alpha J_1 = \frac{\mathrm{d}(\alpha I)}{\mathrm{d}a} = \sqrt{\pi}\,\frac{e^a\operatorname{erf}\sqrt{a}}{\sqrt{a}}, \quad \alpha J_2 = \frac{\mathrm{d}(\alpha J_1)}{\mathrm{d}a} = \alpha J_1\left(1-\frac{1}{2a}\right) + \frac{1}{a}. \tag{7}$$

(erf: Ref. [4, Eq. (7.2.1)]; D§4.3.) Limits: $C_v^{\mathrm{cl}} \to k_B$ as $T \to 0$ (harmonic bottom, equipartition [5, p. 95]) and $C_v^{\mathrm{cl}} \to k_B/2$ as $T \to \infty$. Check: equals direct quadrature of $Z_{\mathrm{cl}}$ to $10^{-9}k_B$.

## 5 Result

| $\alpha L$ | 70 | 100 | 400 | 1000 |
|---|---|---|---|---|
| $\max(C_v^{\mathrm{q}} - C_v^{\mathrm{cl}})/k_B$ | −0.04 (none) | +0.34 | +2.59 | +4.80 |
| at $k_BT/(\hbar^2\alpha^2/2m)$ | – | 0.21 | 0.18 | 0.17 |
| window $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$ | – | [0.18, 0.24] | [0.13, 0.22] | [0.11, 0.20] |

The window opens between $\alpha L = 70$ and 80 ($+0.10k_B$ at 80). The empty box alone gives $C_v^{\mathrm{q}} - C_v^{\mathrm{cl}} = k_B\, r(1-r)/2(2-r)^2$ with $r = \lambda_{\mathrm{th}}/L$, at most $k_B/16$ (D§5; also Fig. 4.3 of Ref. [5, p. 95]). This is $\le 0.01k_B$ inside the windows above. Mechanism: D§6.

## 6 Homogeneous strokes

(i) $\alpha \to \alpha/\gamma$, $L \to \gamma L$ ($\alpha L$ fixed) $\Rightarrow$ $V(x) \to \gamma^{-2}V(x/\gamma)$. [1, p. 021013-2 Eq. (3), p. 021013-9 Table I], D§7.2

(ii) $\Rightarrow$ $E_n \to E_n/\gamma^2$ for *every* $n$, with $\psi_n \to \gamma^{-1/2}\psi_n(x/\gamma)$. [1, p. 021013-14, App. A], D§7.3
This is homogeneous level scaling, $E_n^c = q\,E_n^h$ with one $q = \gamma^{-2}$ for all $n$. [5, p. 92], [6, p. 2]

(iii) $\Rightarrow$ $Z(\beta) \to Z(\beta/\gamma^2)$ and $C_v(T) \to C_v(\gamma^2 T)$, for $C_v^{\mathrm{q}}$ and $C_v^{\mathrm{cl}}$ alike. D§7.4

(iv) $\Rightarrow$ $W = (q-1)\int_{T_c/q}^{T_h} C_v\,\mathrm{d}T$ and $\eta = 1-q$. [5, pp. 94–96, Eqs. (4.15)–(4.18)], D§7.5
This holds with the quantum and with the classical $C_v$ [6, p. 2, Eq. (2)]: same $\eta$, more work wherever $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

## References


[1] S. Deffner, C. Jarzynski, A. del Campo, Phys. Rev. X **4**, 021013 (2014). Open access: https://doi.org/10.1103/PhysRevX.4.021013 (also in `docs/refs`).

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[4] *NIST Digital Library of Mathematical Functions*; each equation at https://dlmf.nist.gov/+number, e.g. https://dlmf.nist.gov/7.7.E1.

[5] A. Levy, D. Gelbwaser-Klimovsky, in *Thermodynamics in the Quantum Regime*, Fund. Theor. Phys. **195**, 87 (Springer, 2018): https://doi.org/10.1007/978-3-319-99046-0_4 (paywalled; cited pages are book pages, as in `docs/refs`). Free: https://arxiv.org/abs/1803.05586v1, where the cited items are on pp. 5–7 (Eqs. (15)–(18), Fig. 3).

[6] D. Gelbwaser-Klimovsky *et al.*, Phys. Rev. Lett. **120**, 170601 (2018). Cited pages are those of the free arXiv:1705.11180v2 (also in `docs/refs`): https://arxiv.org/abs/1705.11180v2.